

BILBAO AP, Spain

WMO: 080250

Lat: 43.298N

Lon: 2.906W

Elev: 42

StdP: 100.82

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.1	-4.0	2.7	5.3	-2.3	3.1	5.0	12.3	11.9	10.9	12.0	1.8	90	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	32.2	20.8	29.7	20.3	27.7	19.7	22.7	28.7	21.7	26.9	20.9	25.4	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	15.8	24.6	20.1	14.9	23.6	19.3	14.1	22.8	67.2	29.0	63.4	27.1	60.5	25.4	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.3	7.1	DB	-2.6	38.6	1.6	2.7	-3.8	40.5	-4.7	42.1	-5.7	43.6	-6.8	45.6	
			WB	-3.1	24.6	1.7	1.1	-4.3	25.4	-5.3	26.1	-6.2	26.7	-7.4	27.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	9.5	9.6	11.8	13.2	16.1	19.2	21.1	21.5	19.5	17.1	12.4	10.2
	DBStd	5.44	3.34	3.56	3.59	3.31	3.33	3.19	2.58	2.62	2.99	3.73	3.45	3.61
	HDD10.0	178	49	45	21	7	1	0	0	0	0	1	14	42
	HDD18.3	1516	273	243	206	158	85	25	4	2	20	68	180	252
	CDD10.0	2058	34	35	76	103	191	277	344	357	285	222	86	48
	CDD18.3	353	0	0	2	4	17	52	89	101	55	31	2	0
	CDH23.3	2193	0	2	19	48	153	353	502	591	348	172	4	0
	CDH26.7	689	0	0	2	13	51	137	153	191	109	33	0	0
Wind	WSAvg	2.7	2.9	3.1	3.2	3.0	2.9	2.6	2.5	2.3	2.3	2.5	2.8	2.8
Precipitation	PrecAvg	1174	136	126	97	101	77	60	51	52	63	90	174	121
	PrecMax	1650	303	345	215	220	178	240	117	155	188	228	342	245
	PrecMin	906	6	36	12	32	28	14	17	8	10	15	74	11
	PrecStd	184	79	92	58	60	36	53	29	38	37	56	64	68
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.1	25.9	28.2	31.9	35.9	35.1	35.2	33.2	29.9	23.7	20.9
		MCWB	13.1	12.6	14.5	16.2	19.4	21.6	22.1	22.1	20.5	18.6	16.0	12.7
	2%	DB	18.1	19.2	22.6	23.8	27.2	30.2	30.2	31.1	29.9	27.1	21.2	18.9
		MCWB	11.8	11.7	13.5	14.7	17.7	20.3	21.5	21.4	19.9	17.9	14.6	12.5
	5%	DB	16.4	17.2	20.1	21.1	24.1	27.0	27.8	28.2	27.0	25.1	19.7	17.1
		MCWB	11.2	10.9	12.5	14.0	16.7	19.5	20.5	20.8	19.4	17.2	13.9	11.6
	10%	DB	14.9	15.2	18.0	19.0	21.9	24.4	25.9	26.2	24.9	23.2	17.9	15.8
		MCWB	10.2	10.0	11.7	13.1	15.6	18.5	20.0	20.1	18.7	16.4	13.0	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.0	15.8	17.5	20.5	22.9	23.9	24.6	22.5	20.4	17.4	14.5
		MCDB	18.4	19.6	23.3	26.0	28.9	31.9	31.1	28.7	29.3	25.6	21.5	18.0
	2%	WB	12.8	12.7	14.1	15.7	18.6	21.3	22.4	23.0	21.2	19.1	15.6	13.4
		MCDB	16.5	17.1	20.5	22.0	25.0	27.6	27.9	28.3	26.6	24.2	19.3	16.9
	5%	WB	11.9	11.7	13.2	14.6	17.4	20.2	21.4	21.8	20.3	18.2	14.6	12.4
		MCDB	15.2	15.6	18.6	19.8	22.6	25.4	25.9	26.5	25.0	23.2	18.3	15.9
	10%	WB	11.0	10.8	12.3	13.6	16.4	19.2	20.6	20.9	19.5	17.3	13.7	11.5
		MCDB	13.9	14.2	16.9	17.8	20.6	23.4	24.5	24.9	23.5	21.7	17.0	14.9
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	9.3	9.4	9.4	9.2	9.1	9.7	10.1	9.5	7.6	7.6
		MCDBR	9.4	11.8	13.3	13.7	14.5	14.7	13.6	14.0	14.0	12.4	10.1	8.9
		MCWBR	5.4	6.5	6.6	6.9	6.9	6.6	6.0	6.3	6.2	5.7	5.4	5.0
	5% WB	MCDBR	8.2	10.1	11.7	12.1	12.6	12.5	11.4	12.0	12.0	11.0	8.7	7.9
		MCWBR	5.3	6.1	6.5	6.6	6.5	6.0	5.4	5.9	5.8	5.6	5.3	5.1
Clear-Sky Solar Irradiance	taub	0.329	0.343	0.376	0.398	0.410	0.423	0.413	0.411	0.390	0.366	0.344	0.325	
	taud	2.505	2.452	2.363	2.313	2.314	2.314	2.345	2.361	2.412	2.480	2.495	2.514	
	Ebn at Noon	804	852	861	868	864	852	856	846	837	814	777	773	
	Edh at Noon	71	88	108	123	126	127	122	116	102	85	72	65	
All-Sky Solar Radiation	RadAvg	1.40	2.20	3.43	4.55	5.43	5.96	6.02	5.31	4.25	2.76	1.62	1.31	
	RadStd	0.17	0.30	0.43	0.47	0.40	0.40	0.29	0.32	0.21	0.23	0.20	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page